

General Information

Practical Information

Syllabus

1. Introduction; Organizing and displaying data; Summary measures.
2. Probability distributions; discrete distributions: Binomial distribution; Poisson distribution; Geometric and Negative Binomial distributions.
3. Continuous distributions: Exponential distribution; Gamma distribution; Normal distributions; Weibull distributions; QQ-plot.
4. Sampling distribution; Estimation and confidence intervals: Central Limit Theorem; parameter estimation (method of moments); confidence interval for mean and variance; comparison of two means and variances; introduction to hypothesis testing; Chi-Square Tests.
5. Regression and analysis of variance: simple regression and basic ANOVA tests.
6. Practical statistics: the use of Minitab to analyse data and to interpret the results.
7. Design of technical posters.

Learning Outcomes

- Define a number of common distributions, fit them to data and test the goodness of fit
- State and apply the Central Limit Theorem
- Derive confidence intervals for mean and variance of a sample and compare the means and variances of two samples using appropriate hypothesis tests
- Conduct simple regression analysis and basic ANOVA tests and assess the applicability of the models used to the data
- Interpret the results of the statistical techniques taught on this module and present them accurately, succinctly and attractively to both specialists and nonspecialists
- Display technical material on a poster

Assessment

70% Closed Book Examination (2 hours), 30% Coursework which takes the form of a poster and a short technical report. The pass mark is 50%.

References

- Prem S.Mann *Introductory Statistics (Sixth Edition)*
- Anderson, Sweeney, Williams, Freeman, Shoesmith *Statistics for Business and Economics*
- Walpole and Mayers *Probability and Statistics for Engineers and Scientists*
- Hoel *Introduction to Mathematical Statistics*

Software

We allow the exercises to be completed in either R or via packages in Python but other statistical tools can be used. If you are using something other than Python or R for the coursework, please confirm with the course lecturer before starting the work.